

Methods Camp

Day 2

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Welcome

- ▶ **Congratulations** on making it to Day 2 of orientation week!
- ▶ Today, we will introduce some of the **foundations of statistics**.
- ▶ Again, our goal is to help you become comfortable with the basic language of quantitative research that you will see frequently in methods courses and scholarly papers.

What is Probability?

- ▶ **Probability** is a number between 0 and 1 that measures how likely something is to happen.
- ▶ $P(A) = 0$: the event A **never** happens. $P(A) = 1$: it **always** happens.

Example: Two Coin Tosses

What is the probability of getting exactly one head in 2 tosses of a fair coin?

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- Flip a fair coin twice. Each flip is Heads (H) or Tails (T), equally likely.

1st Coin	2nd Coin	Probability	# of Heads
H	H	$1/4$	2
H	T	$1/4$	1
T	H	$1/4$	1
T	T	$1/4$	0

- All four outcomes are equally likely: $P(\text{outcome}) = 1/4$ each.

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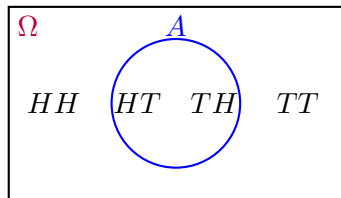
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- All four outcomes are equally likely: $P(\text{outcome}) = 1/4$ each.
- The event “**exactly one head**” contains *two* outcomes (HT, TH) so we **add** their probabilities:

$$P(\text{exactly one head}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

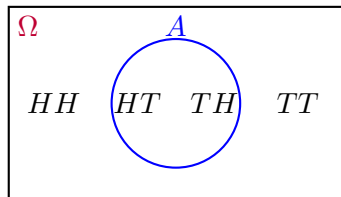
Sample Space and Events

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- ▶ An **event** is a subset of the sample space — one or more outcomes we care about.



Random Variables

What is a Random Variable?

- ▶ A **random variable** is a variable whose value is a numerical outcome of some random process.
- ▶ It assigns a number to each possible outcome of a **sample space**.

Discrete vs. Continuous Random Variables

Discrete

Takes a countable set of values.

$0, 1, 2, 3, \dots$

- ▶ number of children
- ▶ number of arrests
- ▶ voted: 0 or 1

Continuous

Can take any value in an interval.

$2.1, 2.14, 2.141, \dots$

- ▶ income
- ▶ age
- ▶ response time

Random Variable Example: Voting

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$$X = \begin{cases} 1 & \text{if a respondent votes} \\ 0 & \text{if a respondent does not vote} \end{cases}$$

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- ▶ Probabilities range from **0 to 1**.
- ▶ Probabilities across all possible outcomes sum to **1**.

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- ▶ The **expectation** (or **expected value**) of a random variable is its long-run average value, weighted by probability.
- ▶ For a discrete random variable X with possible values $x_1, x_2, \dots$:

$$E[X] = \sum_i x_i P(X = x_i)$$

Expectations of Random Variables

Now going back to the voting example:

$$X = \begin{cases} 1 & \text{if a respondent votes} \\ 0 & \text{if a respondent does not vote} \end{cases}$$

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What is the expected value of X?

Expectations of Random Variables

$$E[X] = (1)(0.70) + (0)(0.30) = 0.70$$

Properties of Expectation

- ▶ Expectation is **linear**: for constants a, b ,

$$E[aX + b] = aE[X] + b$$

- ▶ Expectation of a sum is the sum of expectations:

$$E[X + Y] = E[X] + E[Y]$$

Exercise 1: Random Variables & Expectation

A survey asks respondents how many times they attended religious services last month. Suppose the distribution is:

Times attended (x)	0	1	2	4
$P(X = x)$	0.4	0.3	0.2	0.1

- (a)** Is X discrete or continuous? Why?
- (b)** Compute $E[X]$ by hand.
- (c)** In your own words, what does $E[X]$ tell us?
- (d)** Bonus: if attendance is recoded as $Y = 2X + 1$, what is $E[Y]$? (Use the linearity property.)

Exercise 1: Solutions

(a) X is **discrete**: it only takes on a **countable** set of values $(0, 1, 2, 4)$, not any value in a continuous interval.

(b)

$$E[X] = 0(0.4) + 1(0.3) + 2(0.2) + 4(0.1) = 0 + 0.3 + 0.4 + 0.4 = 1.1$$

(c) $E[X] = 1.1$ is the **long-run average** across *many* respondents — if we surveyed thousands of people and averaged their attendance, we'd land close to 1.1.

(d) Using linearity, $E[aX + b] = aE[X] + b$:

$$E[Y] = E[2X + 1] = 2E[X] + 1 = 2(1.1) + 1 = 3.2$$

Distributions

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2. How likely is each value or range of values?
3. Where are values centered, and how spread out are they?

What is a Distribution?

- ▶ A **probability distribution** describes the possible values of a random variable and their probabilities.
- ▶ Two key tools describe a distribution:
 - ▶ The **PDF** (probability density/mass function)
 - ▶ The **CDF** (cumulative distribution function)

PMF and PDF

Discrete: PMF

The **probability mass function** gives

$$\Pr(X = x).$$

Bars can have positive probability.

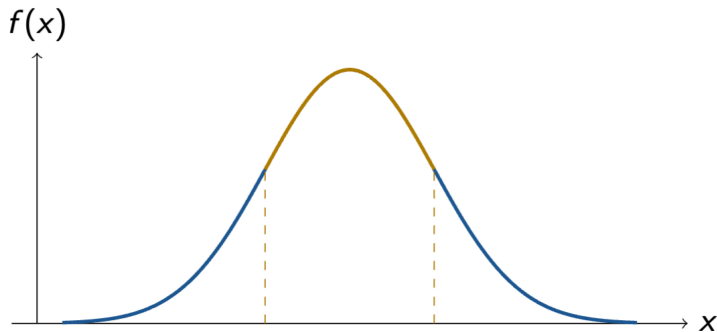
Continuous: PDF

The **probability density function** $f(x)$ describes density.

$$\Pr(a < X < b) = \int_a^b f(x) dx$$

Areas represent probabilities.

PDF: Probability is Area

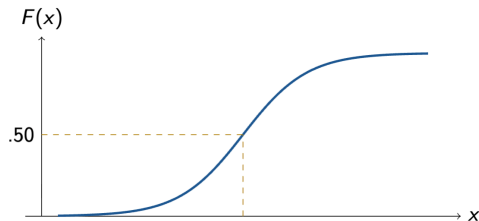


$\Pr(-1 < X < 1)$ is the **area under the gold segment**.

CDF

$$F(x) = \Pr(X \leq x)$$

The **cumulative distribution function** gives the probability of a value at or below x .



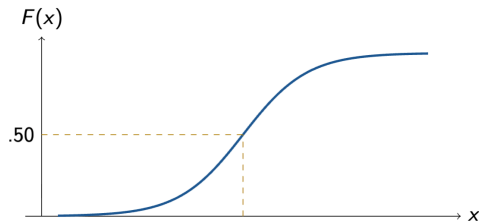
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 $F(0) = .50$.

CDF

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The **cumulative distribution function** gives the probability of a value at or below x .

- ▶ $F(x)$ never decreases.
- ▶ It begins near 0 and approaches 1.
- ▶ It works for both discrete and continuous variables.



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Variance and Standard Deviation

- **Variance** measures how spread out a distribution is around its mean:

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- **Standard deviation** is just the square root of variance:

$$SD(X) = \sqrt{\text{Var}(X)}$$

Mean and Variance Describe Different Features

- ▶ The **mean** describes the **center** of a distribution.
- ▶ The **variance** and **standard deviation** describe how widely values are **spread around the mean**.

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Two distributions can have the same mean but very different amounts of variation.

Common Distributions: Bernoulli

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Examples: employed/not employed; voted/did not vote

Mean: $E[X] = p$

Variance: $V(X) = p(1 - p)$

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- ▶ Example: number of voters among 20 independently sampled eligible residents.

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- ▶ p : probability of success on each trial
- ▶ X : number of successes

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$$E[X] = np, \quad V(X) = np(1 - p)$$

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- ▶ Example: number of protests a city experiences in a year.

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- ▶ $X \sim \text{Poisson}(\lambda)$, where λ is the average rate.

$$E[X] = \lambda, \quad V(X) = \lambda$$

Common Distributions: Normal

- ▶ The **Normal (Gaussian) distribution** is the familiar bell curve, used constantly for **continuous** variables.
- ▶ $X \sim N(\mu, \sigma^2)$, fully described by two parameters:
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 - ▶ σ^2 = variance
- ▶ Many statistical tests **assume** normality.

Exercise 2: Distributions

(a) For each scenario below, name the **most appropriate distribution** (Bernoulli, Binomial, Poisson, or Normal) and briefly justify your choice:

- i. Whether a single respondent owns a home (yes/no)
- ii. The number of home-owners among 200 randomly sampled respondents
- iii. The number of hate-crime incidents reported in a city per month
- iv. Adult height in a large population

(b) For scenario (ii), if $p = 0.65$, what is $E[X]$?

Exercise 2: Solutions

(a)

- i. **Bernoulli** — a single yes/no trial for one respondent.
- ii. **Binomial** — the count of “successes” (home-owners) across $n = 200$ independent Bernoulli trials, each with the same probability p .
- iii. **Poisson** — a count of relatively rare events occurring over a fixed interval (a month), without a natural fixed number of “trials.”
- iv. **Normal** — a continuous measurement that tends to cluster symmetrically around an average, the classic bell-curve shape.

(b) Using $E[X] = np$ for a Binomial:

$$E[X] = 200 \times 0.65 = 130$$

Regression

Why Regression is the Core Tool

- ▶ Regression is how sociologists ask: **"How does Y change as X changes, holding other things constant?"**
- ▶ Many empirical papers you read will use some form of regression.

The Basic Linear Regression Model

A simple regression model relates an outcome Y to a predictor X :

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Y_i : outcome for observation i (person, country, year, etc)

X_i : predictor for observation i

β_0 : intercept

β_1 : slope

ε_i : error term for observation i

The Basic Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

β_0 and β_1 are **fixed for the whole population**.

Y_i , X_i , ε_i vary by observation.

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Individual observations can lie **above or below** that line.

Interpreting the intercept

$$\beta_0 = \mathbb{E}[Y \mid X = 0]$$

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Example:

$$\widehat{\text{wage}} = 8 + 1.5(\text{education})$$

$$\hat{\beta}_0 = 8$$

is the predicted wage when education equals 0.

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The intercept may not be substantively meaningful if $X = 0$ is impossible or far outside the data.

Interpreting the slope

β_1 = change in expected Y for a one-unit increase in X

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For:

$$\widehat{\text{wage}} = 8 + 1.5(\text{education}),$$

one additional year of education is associated with a **\$1.50 increase in predicted hourly wage**, on average.

Error Terms

The **error term** ε_i captures everything that affects Y_i that is **not** explained by X_i .

It may reflect:

- ▶ Omitted factors (things we didn't measure)
- ▶ Measurement error

Assumptions about the Error Term

► **Zero conditional mean:**

$$E[\varepsilon_i \mid X_i] = 0$$

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► **Constant variance:**

$$\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$$

The spread of the errors is constant across values of X .

Assumptions about the Error Term

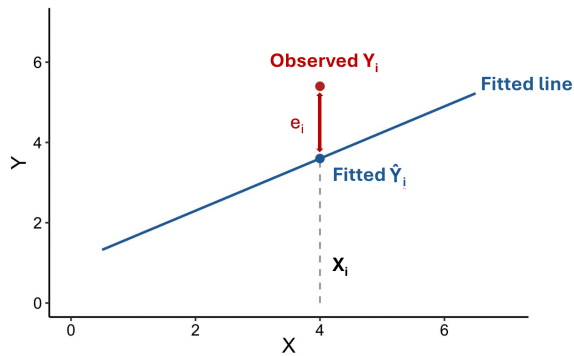
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Assumptions about the Error Term

- ▶ **Independence**: The error for one observation is not systematically related to the error for another observation.
- ▶ **Normality**: The errors are normally distributed at each value of X .

Observed Value, Fitted Value, and Residual

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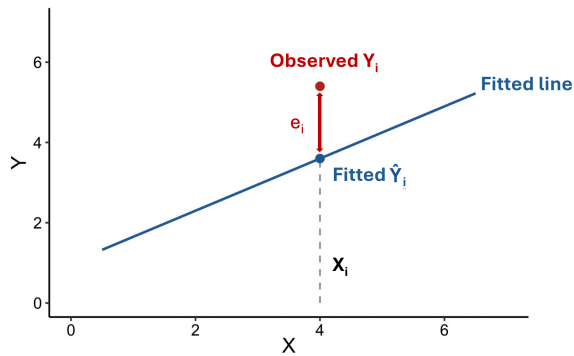


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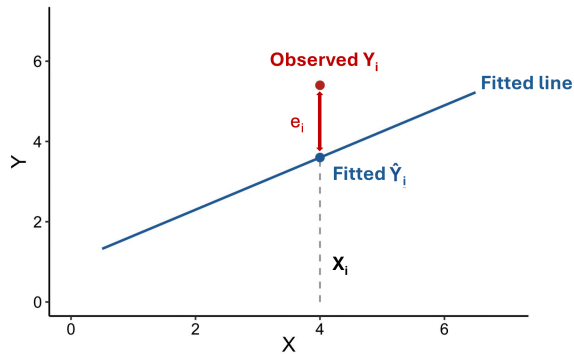
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► $\hat{Y}_i$: **fitted or predicted value**

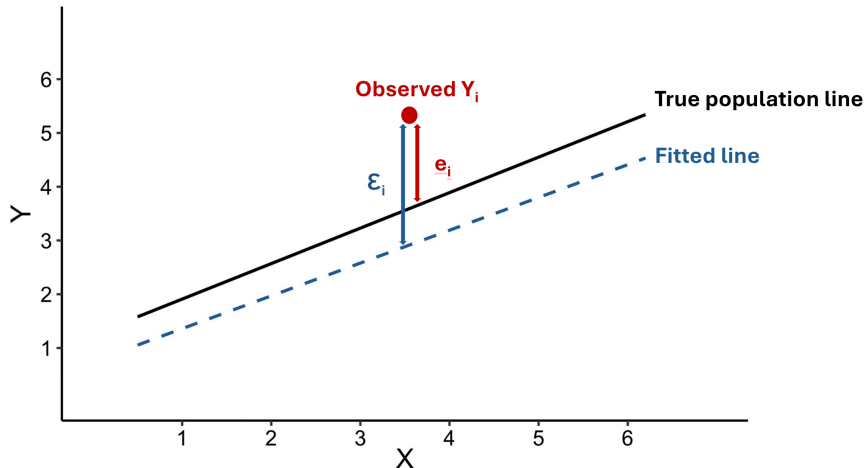
► e_i : **residual**



Error Term vs Residual

Error ε_i	Residual e_i
Part of the population model	Calculated after fitting a sample model
Unobserved	Observed from the fitted model
Distance from the true line	Distance from the estimated line

Error Term vs Residual



Model Example: OLS

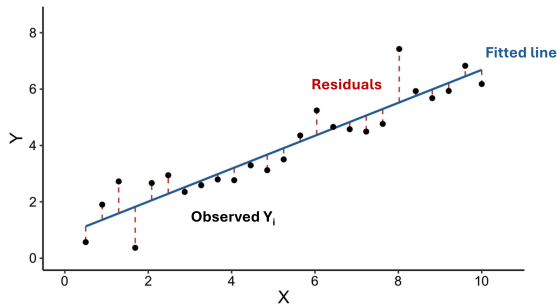
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- ▶ Many possible lines could be drawn through the data.
- ▶ Each possible line produces different **predicted values** and **residuals**.
- ▶ **Ordinary least squares (OLS)** selects the line with the smallest sum of squared residuals.



Estimates, Estimators, and Estimands: Why the Distinction Matters

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- ▶ **Estimator**: the **procedure or formula** used to estimate the estimand from data, such as OLS.
- ▶ **Estimate**: the **numerical result** obtained by applying the estimator to a specific dataset, $\hat{\beta}_1 = 2,300$.

Exercise 3: Regression

Suppose a researcher wants to examine the relationship between parents' years of schooling (X) and their adult child's occupational prestige score (Y) across all U.S. families.

- (a) Write the regression equation using index notation. What does the subscript i represent?
- (b) Interpret the slope coefficient.
- (c) Identify the **estimand**, a plausible **estimator**, and an example of an **estimate**.
- (d) Name one plausible factor that could be included in the **error term**, ε_i .

Exercise 3: Solutions

(a)

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

The subscript i identifies a particular adult child or family.

(b) β_1 represents the expected difference in the child's occupational prestige score associated with one additional year of parental schooling.

(c)

- ▶ **Estimand**: the population slope β_1
- ▶ **Estimator**: ordinary least squares (OLS)
- ▶ **Estimate**: the value obtained from the sample, such as $\hat{\beta}_1 = 1.5$

(d) Parental income, neighborhood conditions, or the local labor market.